THE UNITED REPUBLIC OF TANZANIA
THE PRESIDENT'S OFFICE
REGIONAL ADMINISTRATION AND LOCAL GOVERNMENT
NAMANGA FORM THREE PRE-MIDTERM EXAMINATION

BIOLOGY

MARKING SCHEME AUGUST 2024

SECTION A (16 Marks)

1.

(i)	(ii)	(iii)	(iv)	(v)	(vi)	(vii)	(viii)	(ix)	(x)
E	B	C	C	D	A	A	C	B	D

2. ***01@ = 10 marks***

LIST A	(i)	(ii)	(iii)	(iv)	(v)	(vi)
LIST B	F	G	A	D	C	B

01@ = 06 marks

SECTION B (60 Marks)

3. Dental caries
   - Signs and symptoms include tooth ache, holes in the teeth and bad breath (02 marks)
   - This occurs when bacteria destroy the outer part of the tooth due to the formation of plaque and consequently acids. (03 marks)
   - Control is by brushing ones teeth, using mouth wash; minimize the eating of too sugary foods, having regular dental check up and by removing of affected tooth. (04 marks)

4. (a)
   (i) At trophic levels, species are categorized as autotrophs (or primary producers) example, green plants (1½ marks), *heterotrophs (or consumers)*, and Detritivores (or decomposers), examples bacteria and fungi. ((1½ marks)
   (ii) Heterotrophs can be further sub-divided into different functional groups, including primary consumers (herbivores), secondary consumers (carnivorous predators that feed exclusively on herbivores) and tertiary consumers (predators that feed on a mix of herbivores and predators). (03 marks)

   (b) Because predators are higher up the food chain. In the food chain an organism passes only part of the energy it receives from food. With less energy, each level in the food chain supports fewer individuals than the one below. (03 marks) **(Total: 09 marks)**

5. (a)
   - Amoebic movement: the movement involves streaming of cytoplasm toward peripheral regions of the cell, forming pseudopodia
   - The movement involves beating of small hair-like structures called cilia that surround the body of the organism.

- The movement involves whipping of the whip-like structure called flagellum which is located on one end of the organism (@1 = 03 marks)

(b)

- Protections of the internal organs
- Facilitate movement/locomotion
- Storage of minerals
- Formation of red blood cells
- Giving shape to the body
- Provide support of the body (@1 = 06 marks)

6. Natural classification groups organisms by using homologous features. Homologous features are ones that are similar in structure and having the same evolutionary origin. An example of homologous feature is the wing of the bird, bat and the arms of the human. (02 marks)

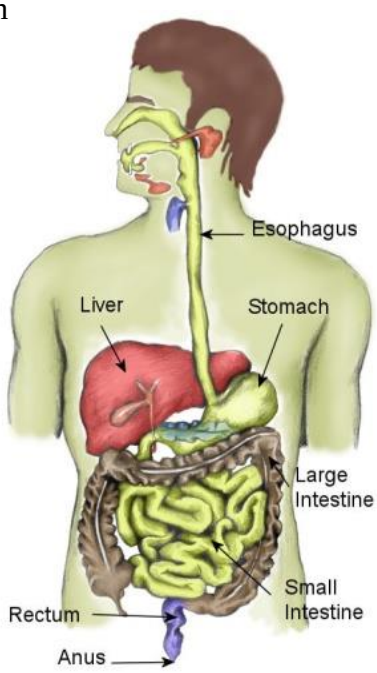
Advantages of Natural classification

- It is accurate enough as it uses many features
- It is not biased (Any four 04 marks)
- Ends up with related organisms being put in the same group or vice versa
- Its universal and hence facilitating communication about organism all over the world
- Avoids confusion: Unrelated organisms can't be placed in the same group.

Disadvantages of Natural classification

- It is not based on easy and simple characteristics but on evolutionary relationships.
- It is time consuming
- It is expensive
- It requires more research (Any three 03 marks)

7. (a) Human digestive system



(Title: 00½ marks
Diagram: 1½ Marks
Label: 6 × 00½ = 3 Marks)

(b) The nutrients digested: Starch and protein (lipid may be present, marks awarded for starch and protein only)

- Starch:
  Enzymes: salivary amylase/ ptyalin , which converts starch into maltose, in the mouth, then maltase converts maltose into glucose, in the ileum, final end product: glucose (02 marks)
- Protein:
  Enzymes: pepsin, which converts protein into peptides, in the stomach (trypsin also convert proteins into peptides in the duodenum), and then peptidase convert peptides into amino acids, in the ileum, final product: amino acids (02 marks)

8.

The pinna collects and directs the sound waves to the ear canal which directs the waves to the ear drum/tympanic membrane. The ear drum vibrates hence converting sound waves into vibrations which are transmitted to ear ossicles behind the ear drum in the middle ear. (03 marks)

The ossicles (bones) are malleus (hammer) which is directly attached to the ear drum, incus (anvil) which is connected to the malleus, and the stapes (stirrup) which is connected to the malleus and to the oval window on the other end, the oval window is a membrane that separates the middle ear and the inner ear. In the middle ear the vibration is amplified 20 times. The vibration of the ossicles in the oval window set up vibration in the liquid of the inner ear and cochlea. (02 marks)

In the inner ear, the vibrations of the endolymph stimulate the organ of corti, which sends the impulse to the brain via the auditory nerve. (02 marks)

The brain is able to sense and interpret different sound volume, pitch, source, and loudness. (02 marks)

SECTION C (30 Marks)

9. Division Coniferophyta or the conifers belong to Kingdom Plantae. These plants are gymnosperms meaning non-flowering. They are non-flowering but seed bearing plants. They have vascular tissues. They are woody plants majority being evergreen trees. Few varieties are shrubs *1½ marks*

Distinctive characteristics

- Conifers are non-flowering but seed producing plants.
- They have seeds born in cones shaped structures rather than in fruits.
- They are gymnosperms which mean naked-seed plants because they produce seeds in cones rather than inside an embryo. *@1½ = 04½ marks*
- The leaves are needle-shaped and have a thick cuticle for protection and to decrease water loss.

Advantages of Conifers

- Are very important for our economy because of their softwood, which is used for paper and timber, as well as cedar, which many people use for their homes because of its beauty and resistance to insects. *@1½ = 07½ mark*
- Decoration and ceremony: Many pine species make attractive ornamentals. They are planted in parks and larger gardens.

- As medicine: Pines and cypress are rich in vitamin; their branches cones and oils can be extracted and used as nutrient supplements.
- Due to its soft texture conifers are widely used to manufacture Play wood.
- As other higher plants conifers are source of heat energy, used as firewood and charcoal.
- Thick forests of conifers form green belts that modify the climate.
- Conifers form a large arena for Biological research Ecological studies.
- Food uses: Some species have large seeds, called pine nuts that are harvested and sold for cooking and baking. The soft, moist, white inner bark (cambium) found clinging to the woody outer bark is edible and very high in vitamins A and C.
- Conclusion: *1½ marks*

Total: 15 marks

10.

The adaptations of the parts of the heart to their functions

- The heart has muscles that are never fatigued.
- The heart has Sino Atrial Node that act as a pacemaker by regulating the heartbeat rates and excitation of the heart.
- There are coronary arteries which lay over the heart surface to provide the heart muscle itself with oxygenated blood.
- Has muscular ventricles, with the left ventricle being more muscular to facilitate pumping of the oxygenated blood around the body.
- There are tricuspid and bicuspid valves between the atria and the ventricles, which ensure the blood flows upwards into the major arteries and not back into the atria.
- At the base of the major arteries, there are semilunar valves which prevent the blood returning to the heart as the ventricles relax. Any $5 \times 03 = 15$ marks
- There is pericardium (pericardial membrane) around the heart that produces lubricating fluid and supports the heart.

Total: 15 marks

11. Natural barriers and the immune system

Natural barrier and the immune system defend the body against infections and diseases.

Natural barrier form the first line of defense. The components of natural barrier include the skins, mucous membrane, cilia/hair, ear wax, sweat, tears, saliva, stomach acids and urine flow. *01½ marks*

- The skin forms a physical barrier to pathogen by preventing them from entering inside of the body.
- The mucous membrane also prevents pathogen from passing through into the tissues and organs.
- The cilia are found along the respiratory tract and trap and prevent pathogen and dust particles from entering the system.
- Earwax traps and therefore prevents dust and pathogen from entering the body through the ear.
- Tears have antiseptic substance that kills germs and therefore preventing them from entering the body.
- Saliva has enzymes that can destroy pathogen and therefore preventing them from entering the body.
- Stomach acid is the acid found in the stomach; it is hydrochloric acid. The acid kill germs that have managed to enter in the stomach with food. Any $6 \times 01 = 06$ marks
- Urine flow flush out pathogens from the urinary bladder.

The components of the immune system include white blood cells, lymph nodes, spleen, liver, bone marrow and tonsils and the thymus. *01½ marks*

- The white blood cells attack and destroy pathogens.
- The lymph nodes trap pathogens which are then destroyed by white blood cells in the lymph.
- Spleen filters blood by removing broken down cells.
- The liver keeps the body safe by removing toxins.
- The bone marrow produces the white blood cells.
- Tonsils and thymus also produce antibodies.

Any $6 \times 01 = 06$ marks

Total: 15 marks